

Mark schemes

Q1.

$$PI: y_{PI} = kx^2e^x$$

M1

$$y'_{PI} = 2kxe^x + kx^2e^x$$

$$y''_{PI} = 2ke^x + 4kxe^x + kx^2e^x$$

Product rule used in finding both derivatives

m1

$$2ke^x + 4kxe^x + kx^2e^x - 4kxe^x - 2kx^2e^x + kx^2e^x = 6e^x$$

Subst. into DE

m1

$$2k = 6; k = 3; y_{PI} = 3x^2e^x$$

CSO

A1

$$(GS: y =) e^x(Ax + B) + 3x^2e^x$$

$e^x(Ax + B) + kx^2e^x$, ft c's k.

B1F

[5]

Q2.

$$\text{Solving } \frac{d^2y}{dt^2} - 6\frac{dy}{dt} + 10y = e^{2t} (*)$$

$$\text{Auxl. Eqn. } m^2 - 6m + 10 = 0$$

$$(m - 3)^2 + 1 = 0$$

PI Completing sq or using quadratic formula to find m.

M1

$$m = 3 \pm i$$

A1

$$CF (y_{CF} =) e^{3t}(A \cos t + B \sin t)$$

*OE Condone x for t here; ft c's 2 **non-real** values for 'm'.*

M1

$$\text{For PI try } (y_{PI} =) ke^{2t}$$

Condone x for t here

M1

$$4k - 12k + 10k = 1 \Rightarrow k = \frac{1}{2}$$

A1

GS of (*) is ($y_{GS} =$) $e^{3t}(A \cos t + B \sin t) + \frac{1}{2}e^{2t}$
CF + PI with 2 arb. constants and both CF and PI functions of t only

B1F

[6]

Q3.

(a) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y = 3x - 8e^{-3x}$

P. Integral: $y \text{ PI} = a + bx + cxe^{-3x}$

$$y' \text{ PI} = b + ce^{-3x} - 3cxe^{-3x}$$

Product rule used at least once giving terms in the form
 $\pm pe^{-3x} \pm qxe^{-3x}$

M1

$$y'' \text{ PI} = -6ce^{-3x} + 9cxe^{-3x}$$

$$-6ce^{-3x} + 9cxe^{-3x} + 2b + 2ce^{-3x} - 6cxe^{-3x}$$

$$-3a - 3bx - 3cxe^{-3x} = 3x - 8e^{-3x}$$

Substitution into LHS of DE

M1

$$-3b = 3; 2b - 3a = 0; -4c = -8$$

Dep on 2nd M only Equating coeffs to obtain at least two of these correct eqns; PI by correct values for at least two constants

m1

$$b = -1; c = 2; a = -\frac{2}{3}$$

Dep on M1M1m1 all awarded A1 if any two correct;
A2 if all three correct but do not award the 2nd A mark
if terms in xe^{-3x} were incorrect in the M1 line

A2,1,0

$$[y_{PI} = -\frac{2}{3} - x + 2xe^{-3x}]$$

5

(b) Aux. eqn. $m^2 + 2m - 3 = 0$
 $(m + 3)(m - 1) = 0$

*Factorising or using quadratic formula OE PI by correct
two values of 'm' seen / used*

M1

$$(y_{CF} =) Ae^{-3x} + Be^x$$

A1

$$(y_{GS} =) Ae^{-3x} + Be^x - \frac{2}{3} - x + 2xe^{-3x}$$

*c's CF + c's PI with 2 arbitrary constants, non-zero
values for a, b and c and no trig or ln terms in c's CF*

B1F

3

(c) $x = 0, y = 1 \Rightarrow 1 = A + B - \frac{2}{3}$

Only ft if previous B1F has been awarded

B1F

$$\frac{dy}{dx} = -3Ae^{-3x} + Be^x - 1 + 2e^{-3x} - 6xe^{-3x}$$

As $x \rightarrow \infty$, ($e^{-3x} \rightarrow 0$ and) $xe^{-3x} \rightarrow 0$

Must treat xe^{-3x} separately

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(As $x \rightarrow \infty$, $\frac{dy}{dx} \rightarrow -1$ so) $B = 0$

$B = 0$, where B is the coefficient of e^x .

B1

When $B = 0$, $1 = A - \frac{2}{3} \Rightarrow A = \frac{5}{3}$

$$y = \frac{5}{3} e^{-3x} - \frac{2}{3} - x + 2xe^{-3x}$$

A1

4

[12]

Q4.

$$m^2 + 2m + 10 = 0$$

PI

M1

$$m = -1 \pm 3i$$

A1

Complementary function is $(y =) e^{-x} (A \cos 3x + B \sin 3x)$

*OE Ft on incorrect **complex value** of m*

A1F

Particular integral: try $y = ke^x$

M1

$$k + 2k + 10k = 26 \Rightarrow k = 2$$

A1

$$(GS \ y =) e^{-x} (A \cos 3x + B \sin 3x) + 2e^x$$

c's CF + c's non-zero PI but must have 2 arb consts

B1F

$$x = 0, y = 5 \Rightarrow 5 = A + 2 \text{ so } A = 3$$

ft c's k ie $A = 5 - k, k \neq 0$

$$\frac{dy}{dx} =$$

*Attempt to differentiate c's **GS***

$$e^{-x} (-3A \sin 3x + 3B \cos 3x - A \cos 3x - B \sin 3x) + 2e^x$$

(ie CF + PI)

$$11 = 3B - A + 2 \quad (B = 4)$$

A1

$$y = e^{-x} (3 \cos 3x + 4 \sin 3x) + 2e^x$$

CSO

A1

[10]

Q5.

(a) $u = xy$

$$\frac{du}{dx} = y + x \frac{dy}{dx}$$

Product rule OE

M1

OE

A1

$$\frac{d^2u}{dx^2} = \frac{dy}{dx} + \left(\frac{dy}{dx} + x \frac{d^2y}{dx^2} \right)$$

OE

A1

$$x \frac{d^2y}{dx^2} + 2(3x + 1) \frac{dy}{dx} + 3y(3x + 2) = 18x$$

$$\left(x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} \right) + 6 \left(x \frac{dy}{dx} + y \right) + 9xy = 18x$$

$$\frac{d^2u}{dx^2} + 6 \frac{du}{dx} + 9u = 18x$$

CSO AG Be convinced

A1

4

(b) $\frac{d^2u}{dx^2} + 6 \frac{du}{dx} + 9u = 18x$

CF: Aux eqn $m^2 + 6m + 9 = 0$

PI

M1

$(m + 3)^2 = 0$ so $m = -3$

PI

A1

CF: $(u =) e^{-3x} (Ax + B)$

A1F

PI: Try $(u =) px + q$

$0 + 6p + 9(px + q) = 18x$

PI. Must be more than just stated

M1

$9p = 18, 6p + 9q = 0$

m1

$$p = 2; \quad q = -\frac{12}{9}$$

Both

A1

$$u = e^{-3x} (Ax + B) + 2x - \frac{4}{3}$$

c's CF + c's PI but must have 2 constants, also must be in the form $u = f(x)$

B1F

$$xy = e^{-3x} (Ax + B) + 2x - \frac{4}{3}$$

$$y = \frac{1}{x} \{ e^{-3x} (Ax + B) + 2x - \frac{4}{3} \}$$

A1

8

[12]

Q6.

(a) PI: $y_{PI} = p + qxe^{-2x}$

$$y'_{PI} = qe^{-2x} - 2qxe^{-2x}$$

Product rule used

M1

$$y''_{PI} = -4qe^{-2x} + 4qxe^{-2x}$$

$$-4q e^{-2x} + 4qxe^{-2x} + qe^{-2x} - 2qxe^{-2x}$$

$$-2p - 2qxe^{-2x} = 4 - 9e^{-2x}$$

Subst. into DE

M1

$$-3q = -9 \quad \text{and} \quad -2p = 4$$

Equating coefficients

m1

$$-3q = -9 \text{ so } q = 3;$$

A1

$$-2p = 4 \text{ so } p = -2;$$

$$[y_{PI} = 3x e^{-2x} - 2]$$

B1

5

(b) Aux. eqn. $m^2 + m - 2 = 0$

$$(m - 1)(m + 2) = 0$$

Factorising or using quadratic formula OE

PI by correct two values of 'm' seen/used

M1

$$y_{CF} = Ae^x + Be^{-2x}$$

A1

$$y_{GS} = Ae^x + Be^{-2x} + 3xe^{-2x} - 2$$

(y_{GS}) = c's CF + c's PI, provided 2 arbitrary constants

B1F

3

(c) $x = 0, y = 4 \Rightarrow 4 = A + B - 2$

Only ft if exponentials in GS

B1F

$$\frac{dy}{dx} = Ae^x - 2Be^{-2x} + 3e^{-2x} - 6xe^{-2x}$$

As $x \rightarrow \infty$, ($e^{-2x} \rightarrow 0$ and) $xe^{-2x} \rightarrow 0$

E1

As $x \rightarrow \infty$, $\frac{dy}{dx} \rightarrow 0$ so $A = 0$

B1

When $A = 0$, $4 = 0 + B - 2 \Rightarrow B = 6$

$$y = 6e^{-2x} + 3xe^{-2x} - 2$$

$$y = 6e^{-2x} + 3xe^{-2x} - 2 \quad \text{OE}$$

B1

4

[12]

Q7.

(a) $y_{PI} = pxe^{-2x} \Rightarrow \frac{dy}{dx} = pe^{-2x} - 2pxe^{-2x}$

Product Rule used

M1

$$\Rightarrow \frac{d^2 y}{dx^2} = -2pe^{-2x} - pe^{-2x} + 4pxe^{-2x}$$

A1

$$-4pe^{-2x} + 4pxe^{-2x} + 3pe^{-2x} - 6pxe^{-2x} + 2pxe^{-2x} = 2e^{-2x}.$$

Sub. into DE

M1

$$-pe^{-2x} = 2e^{-2x} \Rightarrow p = -2$$

ft one slip in differentiation

A1F

4

(b) Aux. eqn. $m^2 + 3m + 2 = 0$

$$\Rightarrow m = -1, -2$$

B1

CF is $Ae^{-x} + Be^{-2x}$

ft on real values of m only

M1

GS $y = Ae^{-x} + Be^{-2x} - 2xe^{-2x}$.

Their CF + their PI must have 2 arb consts

B1F

When $x = 0, y = 2 \Rightarrow A + B = 2$

Must be using GS; ft on wrong non-zero values for p and m

B1F

$$\frac{dy}{dx}$$

$$= -Ae^{-x} - 2Be^{-2x} - 2e^{-2x} + 4xe^{-2x}$$

Must be using GS; ft on wrong non-zero values for p and m

B1F

$$\frac{dy}{dx}$$

When $x = 0, \frac{dy}{dx} = 0 \Rightarrow -A - 2B - 2 = 0$

Must be using GS; ft on wrong non-zero values for p and m and slips in finding $y'(x)$

B1F

Solving simultaneously, 2 eqns each in two arbitrary constants

m1

$$A = 6, B = -4; y = 6e^{-x} - 4e^{-2x} - 2xe^{-2x}.$$

CSO

A1

8

[12]

Q8.

Aux. eqn. $m^2 + 4 = 0 \Rightarrow m = \pm 2i$

B1

CF is $A \cos 2x + B \sin 2x$

OE. If m is real give M0

M1

ft on incorrect complex value for m

A1F

PI: Try $ax^2 + b + c \sin x$

Award even if extra terms, provided the relevant coefficients are shown to be zero.

M1M1

$$2a - c \sin x + 4ax^2 + 4b + 4c \sin x = 8x^2 + 9 \sin x$$

$$a = 2, b = -1,$$

Dep on relevant M mark

A1

$$c = 3$$

Dep on relevant M mark

A1

$$(y =) A \cos 2x + B \sin 2x + 2x^2 - 1 + 3 \sin x$$

Their CF + their PI. Must be exactly two arbitrary constants

B1F

[8]

Q9.

(a) $-4k \sin 2x + k \sin 2x = \sin 2x$



M1A1

$$k = -\frac{1}{3}$$

Accept correct PI

A1

3

(b) (Aux. eqn $m^2 + 1 = 0$) $m = \pm i$
PI

B1

CF: $A \cos x + B \sin x$

M0 if m is real

M1

OE Ft on incorrect complex values for m
For the A1F do not accept if left in the
form $Ae^{ix} + Be^{-ix}$

A1F

(GS: $y =$) $A \cos x + B \sin x - \frac{1}{3} \sin 2x$
c's CF + c's PI but must have 2 constants

B1F

EXAM PAPERS PRACTICE [7]

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Q10.

(a) $-k \sin x + 2k \cos x + 5k \sin x = 8 \sin x + 4 \cos x$
Differentiation and subst. into DE

M1A1

$$k = 2$$

A1

3

(b) Auxl eqn $m^2 + 2m + 5 = 0$
 $m = \frac{-2 \pm \sqrt{4 - 20}}{2}$

Formula or completing sq. PI

M1

$$m = -1 \pm 2i$$

A1

$$\text{CF: } \{y_C\} = e^{-x}(A \sin 2x + B \cos 2x)$$

ft provided m is not real

A1F

$$\text{GS } \{y\} = e^{-x}(A \sin 2x + B \cos 2x) + k \sin x$$

ft on CF + PI; must have 2 arb consts

B1F

$$\text{When } x = 0, y = 1 \Rightarrow B = 1$$

B1F

$$\begin{aligned} \frac{dy}{dx} &= -e^{-x}(A \sin 2x + B \cos 2x) \\ &\quad + e^{-x}(2A \cos 2x - 2B \sin 2x) + k \cos x \end{aligned}$$

Product rule

M1

$$\text{When } x = 0, \frac{dy}{dx} = 4 \Rightarrow 4 = -B + 2A + k$$

PI

A1

$$\Rightarrow A = \frac{3}{2}$$

$$y = e^{-x} \left(\frac{3}{2} \sin 2x + \cos 2x \right) + 2 \sin x$$

CSO

A1

8

[11]

Q11.

(a) $(m + 2)^2 = -1$

Completing sq or formula

M1

$$m = -2 \pm i$$

A1

CF is $e^{-2x} (A \cos x + B \sin x)$

If m is real give M0

M1

{or $e^{-x} A \cos (x + B)$

but not $Ae^{(-1+i)x} + Be^{(-1-i)x}$ }

Ft on wrong a's and b's but roots must be complex

A1ft

PI try $y = p \Rightarrow 5p = 5$ PI is $y = 1$

B1

GS $y = e^{-2x} (A \cos x + B \sin x) + 1$

Their CF + their PI with two arbitrary constants.

B1ft

6

(b) $x = 0, y = 2 \Rightarrow A = 1$

Provided previous B1ft awarded

B1ft

$y'(x) = -2e^{-2x} (A \cos x + B \sin x) + e^{-2x} (-A \sin x + B \cos x)$

Product rule used

M1A1ft

$y'(0) = 3 \Rightarrow 3 = -2A + B \Rightarrow B = 5$

Ft on one slip

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A1ft

$y = e^{-2x} (\cos x + 5 \sin x) + 1$

4

[10]

Q12.

(a) Aux eqn: $m^2 - 2m - 3 = 0$

M1

$m = -1, 3$

PI

A1

CF ($y_C =$) $Ae^{3x} + Be^{-x}$

M1

Try ($y_{PI} = a e^{-2x} + b$)

M1

$$\frac{dy}{dx} = -2ae^{-2x}$$

A1

$$\frac{d^2y}{dx^2} = 4ae^{-2x}$$

A1

Substitute into DE gives

$$4ae^{-2x} + 4ae^{-2x} - 3ae^{-2x} - 3b = 10e^{-2x} - 9$$

M1

$$\Rightarrow a = 2$$

A1

$$b = 3$$

B1

$$(y_{GS}) = Ae^{3x} + Be^{-x} + 2e^{-2x} + 3$$

(c's CF+c's PI) with 2 arbitrary constants

B1F

10

(b) $x = 0, y = 7 \Rightarrow 7 = A + B + 2 + 3$

Only fit if exponentials in GS and two arbitrary constants remain
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B1F

$$\frac{dy}{dx} = 3Ae^{3x} - Be^{-x} - 4e^{-2x}$$

As $x \rightarrow \infty, e^{-kx} \rightarrow 0, \frac{dy}{dx} \rightarrow 0$ so $A = 0$

B1

When $A = 0, 5 = 0 + B + 3 \Rightarrow B = 2$

Must be using 'A' = 0

B1F

$$y = 2e^{-x} + 2e^{-2x} + 3$$

CSO

Q13.

Auxl. eqn $m^2 - 4m + 3 = 0$

PI

M1

$m = 3$ and 1

PI

A1

CF is $A e^{3x} + B e^x$

A1F

PI Try $y = a + b \sin x + c \cos x$

Condone 'a' missing here

M1

$y'(x) = b \cos x - c \sin x$

A1

$y''(x) = -b \sin x - c \cos x$

ft can be consistent sign error(s)

A1F

Substitute into DE gives

M1

$a = 2$

B1

$4c + 2b = 5$ and $2c - 4b = 0$

A1

$b = 0.5,$

ft a slip

A1F

$c = 1$

ft a slip

A1F

GS: $y = A e^{3x} + B e^x + 2 + 0.5 \sin x + \cos x$

$y =$ candidate's CF and candidate's PI
(must have exactly two arbitrary constants)

B1F

12

[12]

Q14.

(a) $y_{PI} = kx^2 e^{5x} \Rightarrow y' = 2kx e^{5x} + 5kx^2 e^{5x}$

Product rule to differentiate $x^2 e^{5x}$

M1

A1

$\Rightarrow y'' = 2k e^{5x} + 10kx e^{5x} + 10kx e^{5x} + 25kx^2 e^{5x}$

A1ft

$\Rightarrow 2k e^{5x} + 20kx e^{5x} + 25kx^2 e^{5x} - 10(2kx e^{5x} + 5kx^2 e^{5x}) + 25kx^2 e^{5x} = 6e^{5x}$

Substitution into differential equation

M1

A1

$2k = 6 \Rightarrow k = 3$

Only ft if $x e^{5x}$ and $x^2 e^{5x}$ terms all cancel out

A1ft

6

(b) Aux. eqn. $m^2 - 10m + 25 = 0 \Rightarrow m = 5$

PI

B1

CF is $(A + Bx)e^{5x}x$

M1

Their CF + their / our PI

M1

GS $y = (A + Bx)e^{5x}x + 3x^2 e^{5x}x$

ft only on wrong value of k

A1ft

4

[10]

Q15.

(a) $(m + 1)^2 = -1$

Completing sq or formula

M1

$$m = -1 \pm i$$

A1

2

(b) (i) CF is $e^{-x}(A \cos x + B \sin x)$

If m is real give M0

M1

{or $e^{-x}A \cos(x + B)$ **but not** $Ae^{(-1+i)x} + Be^{(-1-i)x}$ }

ft On wrong a's and b's but roots must be complex.

A1 ✓

{P.Int.} try $y = px + q$

OE

M1

$$2p + 2(px + q) = 4x$$

A1

$$p = 2, \quad q = -2$$

On one slip

A1 ✓

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GS $y = e^{-x}(A \cos x + B \sin x) + 2x - 2$

Their CF + their PI with two arbitrary constants.

B1 ✓

6

(ii) $x = 0, y = 1 \Rightarrow A = 3$

Provided an M1 gained in (b)(i)

B1 ✓

$$y'(x) = -e^{-x}(A \cos x + B \sin x) +$$

Product rule used

M1

$$+ e^{-x}(-A \sin x + B \cos x) + 2$$

A1 ✓

$$y'(0) = 2 \Rightarrow 2 = -A + B + 2 \Rightarrow B = 3$$

Slips

A1 ✓

$$y = 3e^{-x}(\cos x + \sin x) + 2x - 2$$

4

[12]

Q16.

(a) $y = 2x + \sin 2x \Rightarrow y' = 2 + 2\cos 2x$
 $\Rightarrow y'' = -4\sin 2x$

Need to attempt both y' and y''

M1 A1

$$-4\sin 2x - 5(2 + 2\cos 2x) + 4(2x + \sin 2x) =$$

$$8x - 10 - 10\cos 2x$$

CSO AG Substitute. and confirm correct

A1

3

(b) Auxiliary equation $m^2 - 5m + 4 = 0$

M1

$$m = 4 \text{ and } 1$$

A1

CF: $Ae^{4x} + Be^x$

M1

GS: $y = Ae^{4x} + Be^x + 2x + \sin 2x$

Their CF + $2x + \sin 2x$

B1ft

4

(c) $x = 0, y = 2 \Rightarrow 2 = A + B$

Only ft if exponentials in GS

B1ft

$$x = 0, y' = 0 \Rightarrow 0 = 4A + B + 4$$

Only ft if exponentials in GS and differentiated four terms at least

B1ft

Solving the simultaneous equations

M1

gives $A = -2$ and $B = 4$

A1

4

$$y = -2e^{4x} + 4e^x + 2x + \sin 2x$$

[11]

Q17.

(a) $75 \frac{d^2x}{dt^2} = 75g - \frac{450}{12}x - 15 \frac{dx}{dt}$

M1: equation with four terms of the correct format.

A1: Correct terms.

M1A1

A1: Correct signs.

A1

$$75 \frac{d^2x}{dt^2} = 750 - 37.5x - 15 \frac{dx}{dt}$$

$$10 \frac{d^2x}{dt^2} + 2 \frac{dx}{dt} + 5x = 100$$

A1: Correct result from correct working.

A1

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AG

4

(b) CF

$$10\lambda^2 + 2\lambda + 5 = 0$$

M1: Correct auxiliary equation.

M1

$$\lambda = \frac{-2 \pm \sqrt{2^2 - 4 \times 5 \times 10}}{2 \times 10} = -0.1 \pm 0.7i$$

M1: Correct complex solutions.

M1

$$x = e^{-0.1t} (A \sin(0.7t) + B \cos(0.7t))$$

A1: Correct CF

A1

PI

$$x = 20$$

B1: Correct PI.

B1

$$x = e^{-0.1t} (A \sin(0.7t) + B \cos(0.7t)) + 20$$

dM1: Using initial conditions to find B.

dM1

$$x = 0, t = 0 \Rightarrow B = -20$$

A1: Correct value for B.

A1

$$x = e^{-0.1t} (A \sin(0.7t) - 20 \cos(0.7t)) + 20$$

$$\dot{x} = -0.1e^{-0.1t} (A \sin(0.7t) - 14 \cos(0.7t)) +$$

$$e^{-0.1t} (0.7A \cos(0.7t) - 14 \sin(0.7t))$$

dM1: Correct $\dot{x}$

dM1

$$\dot{x} = 12.5, t = 0 \Rightarrow A = 15$$

dM1: Using initial conditions to find A.

dM1

$$x = e^{-0.1t} (15 \sin(0.7t) - 20 \cos(0.7t)) + 20$$

A1: Correct value for A and correct final expression.

A1

10

(c) $v = e^{-0.1t} (12.5 \cos(0.7t) + 12.5 \sin(0.7t))$

$$v = 0$$

M1: Setting derivative equal to zero.

A1: Correct equation including correct derivative.

M1A1

$$\tan(0.7t) = -1$$

dM1: Value for $\tan(0.7t)$.

dM1

A1: Correct value for $\tan(0.7t)$.

A1

$$t = \frac{15\pi}{14} = 3.37 \text{ s}$$

A1: Correct time.

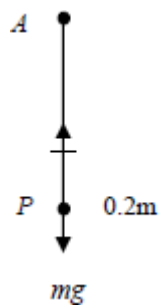
A1

5

[19]

Q18.

(a)



$$T = 0.4g$$

$$T = 0.2k$$

$$0.2k = 0.4g$$

$$k = 2g$$

$$k = 19.6 \text{ Nm}^{-1}$$

M1

A1

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A1F

Both, accept use of λ and l

3

(b) extension = $0.2 + x$

B1

(i) $mg - T = m\ddot{x}$

M1

$$0.4g - 19.6(x + 0.2) = 0.4\ddot{x}$$

ft stiffness

A1F

$$0.4\ddot{x} = -19.6x$$

$$\ddot{x} = -49x \quad (h = -49)$$

A1F

4

ft stiffness

(ii) $\ddot{x} = -\omega^2 x$ *SHM*

(constant < 0)

ft stiffness provided $h < 0$

B1F

1

(iii) $T = \frac{2\pi}{\omega}$

$$T = \frac{2\pi}{7}$$

$T = 0.898 \text{ sec}$
AG

M1

A1

2

(iv) $\max v = a\omega$

$$= 0.1 \times 7$$

$$= 0.7 \text{ ms}^{-1}$$

ft stiffness provided $h < 0$

M1

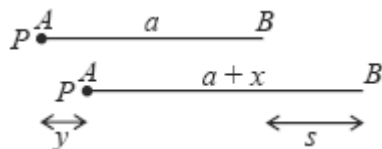
A1F

2

[12]

Q19.

(a)



$$s = \frac{1}{2} at^2$$

B1

$$s = x + y \therefore y = \frac{1}{2}ft^2 - x$$

AG

M1A1

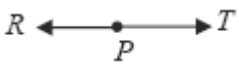
3

(b) $\frac{dy}{dt} = ft - \frac{dx}{dt}$

AG

B1

1

(c)  $\frac{d^2y}{dt^2} = f - \frac{d^2x}{dt^2}$

B1F

$$F = ma \therefore \frac{md^2y}{dt^2} = \frac{2mn^2ax}{a} - 3mn\left(ft - \frac{dx}{dt}\right)$$

M1M1A1

$$f - \frac{d^2x}{dt^2} = 2mn^2x - 3nft + 3n\frac{dx}{dt}$$

$$\frac{d^2x}{dt^2} + 3n\frac{dx}{dt} + 2n^2x = f + 3nft$$

AG

A1

5

(d) A.E. $p^2 + 3p + 2 = 0$
 $(p + 2)(p + 1) = 0$

M1

$$p = -1 \text{ or } p = -2$$

A1

$$x = Ae^{-2t} + Be^{-t} + \frac{f}{4}(6t - 7)$$

M1A1

$$t = 0, x = 0 \Rightarrow 0 = A + B - \frac{7f}{4}$$

$$t = 0, \dot{x} = 0 \Rightarrow \dot{x} = -2Ae^{-2t} - Be^{-t} + \frac{6f}{4}$$

M1

$$0 = -2A - B + \frac{6f}{4}$$

$$B = 2f, A = \frac{-f}{4}$$

A1,A1

$$x = 2fe^{-t} - \frac{f}{4}e^{-2t} + \frac{f}{4}(6t - 7)$$

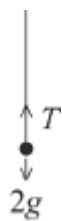
OE

7

[16]

Q20.

(a)



$$2g = \frac{49 \times e}{0.5}$$

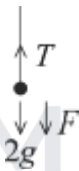
$$e = 0.2 \text{ metres}$$

M1

A1

2

(b)



$$2\ddot{x} = 2g + F - T$$

M1A1

$$2\ddot{x} = 2g + 12 \cos nt - \frac{49(x + 0.2)}{0.5} \quad \text{Tension ft (a)}$$

B1F

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$$2\ddot{x} = 2g + 12 \cos nt - 98x - 19.6$$

ft (a)

A1F

$$\ddot{x} + 49x = 6 \cos nt$$

AG

A1

5

(c) $n = 5$ Pl, $x = A \cos 5t + B \sin 5t$
Accept cos term only

M1

$$\dot{x} = -5A \sin 5t + 5B \cos 5t$$

A1

$$\ddot{x} = -25A \cos 5t - 25B \sin 5t$$

A1

Subs:

$$-25A \cos 5t - 25B \sin 5t + 49A \cos 5t + 49B \sin 5t = 6 \cos 5t$$

$$B = 0, \quad A = \frac{1}{4}$$

A1

A. eqn: $m^2 + 49 = 0 \quad m = \pm 7i$

M1

C.F.: $x = C \cos 7t + D \sin 7t$

A1

Gen sol: $x = C \cos 7t + D \sin 7t + \frac{1}{4} \cos 5t$

M1

$t = 0, x = 0: \quad 0 = C + \frac{1}{4} \quad C = -\frac{1}{4}$

A1

$$\dot{x} = -7C \sin 7t + 7D \cos 7t - \frac{5}{4} \sin 5t$$

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m1

$t = 0, \dot{x} = 0: \quad D = 0$

$$x = \frac{1}{4}(\cos 5t - \cos 7t)$$

A1

10

(d) Resonance occurs when $n = 7$

B1

1

[18]

Q21.

(a) Using $F = ma$,

$$2m\ddot{x} = 2mg - T - 4mnv$$

3 terms at least correct

M1

$$= 2mg - 8mn^2a \cdot \frac{x}{a} - 4mnv$$

A1

$$m\ddot{x} + 4mn^2x + 2mn\dot{x} = mg$$

$$\text{i.e. } \frac{d^2x}{dt^2} + 2n\frac{dx}{dt} + 4n^2x = g$$

A1

3

(b) Substituting $x = Ae^{pt}$

$$p^2 + 2np + 4n^2 = 0$$

M1

$$p = \frac{(-2n \pm \sqrt{4n^2 - 16n^2})}{2}$$

$$= -n \pm \sqrt{3}in$$

A1

C.F. is

$$x = e^{-nt} (A \cos \sqrt{3}nt + B \sin \sqrt{3}nt)$$

Dep M1

B1ft

$$\text{P.I. is } x = \frac{g}{4n^2}$$

B1

General solution is $x =$

$$e^{-nt} (A \cos \sqrt{3}nt + B \sin \sqrt{3}nt) + \frac{g}{4n^2}$$

ft dep M1 first line (b)

M1

At least one of CF, PI correct

A1ft

$$\text{When } t = 0, x = 0, \Rightarrow 0 = A + \frac{g}{4n^2}$$

M1

$$\therefore A = -\frac{g}{4n^2}$$

A1

$$\text{When } t = 0, \frac{dx}{dt} = 0;$$

$$\frac{dx}{dt} = -ne^{-nt}(A\cos\sqrt{3}t + B\sin\sqrt{3}t) +$$

M1

$$e^{-nt}(-\sqrt{3}nA\sin\sqrt{3}nt + \sqrt{3}nB\cos\sqrt{3}nt)$$

A1

$$\Rightarrow 0 = -nA + \sqrt{3}nB$$

$$\therefore B = \frac{1}{\sqrt{3}}A = -\frac{g}{4\sqrt{3}n^2}$$

A1

$$\therefore x = \frac{g}{4n^2} - \left(\frac{g}{4n^2} \cos\sqrt{3}nt + \frac{g}{4\sqrt{3}n^2} \sin\sqrt{3}nt \right) e^{-nt}$$

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A1

2

$$(c) \quad \frac{dx}{dt} = ne^{-nt} \left(\frac{g}{4n^2} \cos\sqrt{3}nt + \frac{g}{4\sqrt{3}n^2} \sin\sqrt{3}nt \right)$$

M1

$$-e^{-nt} \left(-\frac{\sqrt{3}g}{4n^2} \sin\sqrt{3}nt + \frac{g}{4n} \cos\sqrt{3}nt \right)$$

A1

$$\frac{dx}{dt} = 0 \text{ when}$$

$$\cos\sqrt{3}nt + \frac{1}{\sqrt{3}} \sin\sqrt{3}nt + \sqrt{3} \sin\sqrt{3}nt$$

$$- \cos\sqrt{3}nt = 0$$

M1

$$\sin \sqrt{3} \pi t = 0$$

$$\text{First time at rest is when } t = \frac{\pi}{\sqrt{3}\pi}$$

A1

4

[19]

Q22.

(a) $T = 10(x - 0.5)$ or $-10(x - 0.5)$

M1A1

2

(b) $10(x - 0.5) - 2v = 1 \frac{dv}{dt}$
Newton's 2nd law

$$\frac{dv}{dt} + 2v - 10x = -5$$

AG

M1A1F

A1

3

(c) (i) The rate of change of the length of the spring = speed of A relative to B
 OE

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E1

1

$$\therefore \dot{x} = 1 - v$$

(ii) $-\ddot{x} + 2(1 - \dot{x}) - 10x = -5$

M1

$$\ddot{x} + 2\dot{x} + 10x = 7$$

AG

A1

2

(d) Auxiliary equation: $m^2 + 2m + 10 = 0$

M1

$$m = -1 \pm 3i$$

A1

$$\text{CF } x = e^{-t} (A \sin 3t + B \cos 3t)$$

M1

$$\text{PI } x = \alpha$$

$$\dot{x} = \ddot{x} = 0$$

M1

A1F

$$\alpha = \frac{7}{10}$$

$$\text{GS } x = e^{-t} (A \sin 3t + B \cos 3t) + \frac{7}{10}$$

A1F

6

$$(e) \quad t \rightarrow \infty$$

$$e^{-t} \rightarrow 0$$

$$x \rightarrow \frac{7}{10}$$

M1A1F

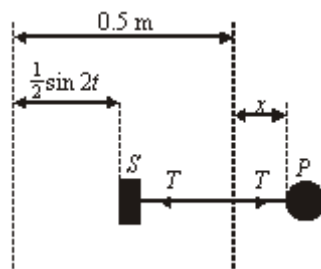
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[16]

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Q23.

(a)



B1

1

$$(b) \quad T = 4.5 \left(\frac{1}{2} \sin 2t - x \right)$$

M1

$$T = 0.5 \ddot{x}$$

A1

$$0.5 \ddot{x} = 4.5 \left(\frac{1}{2} \sin 2t - x \right)$$

Newton's 2nd law

M1

$$\ddot{x} + 9x = \frac{9}{2} \sin 2t$$

AG

A1

4

(c) Auxiliary eqⁿ. $m^2 + 9 = 0$

M1

$$m = \pm 3i$$

A1

C.F.

$$x = A \cos 3t + B \sin 3t$$

B1

For P.I. $x = p \cos 2t + q \sin 2t$

$$\dot{x} = -2p \sin t + 2q \cos 2t$$

$$\ddot{x} = -4p \cos 2t - 4q \sin 2t$$

M1

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$$5p \cos 2t + 5q \sin 2t = \frac{9}{2} \sin 2t$$

M1

$$p = 0, q = \frac{9}{10}$$

For both p and q

A1F

G.S. is

$$x = A \cos 3t + B \sin 3t + \frac{9}{10} \sin 2t$$

A1F

$$x = \dot{x} = 0, t = 0$$

$$\therefore A = 0$$

A1F

$$\dot{x} = 3B \cos 3t + \frac{9}{5} \cos 2t$$

$$\therefore B = -\frac{3}{5}$$

A1F

$$x = -\frac{3}{5} \sin 3t + \frac{9}{10} \sin 2t$$

9

(d) $\dot{x} = -\frac{9}{5} \cos 3t + \frac{9}{5} \cos 2t$

$$= -\frac{9}{5} \cos \frac{6\pi}{5} + \frac{9}{5} \cos \frac{4\pi}{5}$$

$$= 0$$

M1

A1

2

[16]

EXAM PAPERS PRACTICE

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Auxiliary eqⁿ. $m^2 + 121 = 0$
 $m = \pm 11i$
 Or knowing to quote S.H.M

M1

C.F. $x = A \sin 11t + B \cos 11t$

A1

For PI $x = \alpha \sin \omega t + \beta \cos \omega t$

$$\dot{x} = \alpha \omega \cos \omega t - \beta \omega \sin \omega t$$

M1

$$\ddot{x} = -\alpha \omega^2 \sin \omega t - \beta \omega^2 \cos \omega t$$

M1

$$-\alpha\omega^2 \sin \omega t - \beta\omega^2 \cos \omega t + 121 (\alpha \sin \omega t + \beta \cos \omega t) = 3 \sin \omega t$$

$$-\alpha\omega^2 + 121\alpha = 3 \quad \Rightarrow \quad \alpha = \frac{3}{(121 - \omega^2)}$$

$$-\beta\omega^2 + 121\beta = 0 \quad \Rightarrow \quad \beta = 0$$

For α and β

A1F

$$x = A \sin 11t + B \cos 11t + \frac{3}{(121 - \omega^2)} \sin \omega t$$

B1F

[6]

Q25.

(a) Auxiliary equation: $m^2 + 2m + 5 = 0$

M1

$$m = -1 \pm 2i$$

A1

$$\text{CF } \theta = e^{-t}(A \cos 2t + B \sin 2t)$$

A1F

For PI $\theta = \alpha$

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M1

$$\dot{\theta} = 0$$

$$5\alpha = 0.3 \Rightarrow \alpha = 0.06$$

A1

$$\text{GS} = \text{CF} + \text{PI}$$

B1F

6

$$(b) \quad \dot{\theta} = -e^{-t}(A \cos 2t + B \sin 2t) + e^{-t}(-2A \sin 2t + B \cos 2t)$$

Differentiating their GS (must be correct form)

M1 A1F

$$0.2 = e^0 (A \cos 0 + B \sin 0) + 0.06$$

$$A = 0.14$$

ft from part (a)

A1F

$$0 = -e^0(A\cos 0 + B\sin 0) + e^0(-2A\sin 0 + B\cos 0)$$

$$0 = -A + 2B \quad 0 = -0.14 + 2B$$

$$B = 0.07$$

$$\theta = e^{-t}(0.14\cos 2t + 0.007\sin 2t) + 0.06$$

A1F A1F

5

(c) As $t \rightarrow \infty$, $e^{-t} \rightarrow 0$ and

$$e^{-t}(0.14\cos 2t + 0.07\sin 2t) \rightarrow 0$$

Correct from with A or B $\neq 0$

M1

$$\therefore \theta \rightarrow 0.06$$

A1F

2

[13]

Q26.

(a) Substituting $x = Ae^{nt}$ into CF

$$n^2 + 6n + 10 = 0$$

$$n = \frac{-6 \pm \sqrt{36 - 40}}{2}$$

$$-3 \pm i$$

M1

$$\therefore \text{CF is } x = e^{-3t}(A \cos t + B \sin t)$$

M1A1

$$\text{PI: } x = C \sin 3t + D \cos 3t$$

M1 only for PI section if only Csin3t or Dcos3t used

M1

$$\frac{dx}{dt} = 3C \cos 3t - 3D \sin 3t$$

$$\frac{d^2x}{dt^2} = -9C \sin 3t - 9D \cos 3t$$

Substituting into

$$\frac{d^2 x}{dt^2} + 6 \frac{dx}{dt} + 10x = 325 \sin 3t$$

$$\begin{aligned} & -9C \sin 3t - 9D \cos 3t + 18C \cos 3t \\ & -18D \sin 3t + 10C \sin 3t + 10D \cos 3t \end{aligned}$$

M1A1

$$= 325 \sin 3t$$

$$(\sin t) \quad C - 18D = 325$$

$$(\cos t) \quad D + 18C = 0$$

$$325C = 325$$

$$C = 1 \quad \text{B1}$$

$$D = -18$$

B1

$$\therefore \text{PI is } x = \sin 3t - 18 \cos 3t$$

$$x = e^{-3t} (A \cos t + B \sin t) + (\sin 3t - 18 \cos 3t)$$

$$\text{When } x = 0, t = 0$$

$$\Rightarrow 0 = A - 18$$

$$\Rightarrow A = 18$$

B1

$$\frac{dx}{dt} = -3e^{-3t} (A \cos t + B \sin t) +$$

$$e^{-3t} (B \cos t - A \sin t) + 3 \cos 3t + 54 \sin 3t$$

$$\text{When } t = 0, \frac{dx}{dt} = 0$$

$$\Rightarrow 0 = -3A + B + 3$$

$$B = 51$$

B1

$$\therefore x = (18 \cos t + 51 \sin t)e^{-3t} +$$

$$\sin 3t - 18 \cos 3t$$

$$\text{Accept } x = 54e^{-3t} \cos(t - 1.232)$$

$$+ (\sin 3t - 18 \cos 3t)$$

$$\text{allow } 54, 54.1, 54.08$$

B1

11

(b) When t is large, $x \approx \sin 3t - 18 \cos 3t$

B1

This is periodic with period $\frac{2\pi}{3}$

B1

and amplitude $5\sqrt{13}$

Accept $\sqrt{325}$

B1

3

[14]

Q27.

(a) When displacement is x ,

$$\text{Tension} = \frac{\lambda x}{l} = \frac{4mn^2 a x}{2a}$$

$$= 2mn^2 x$$

B1

Using $F = ma$,

$$m = \frac{d^2 x}{dt^2} = mg - 2mn^2 x - 2mn \frac{dx}{dt}$$

M1 A1

$$\frac{d^2 x}{dt^2} + 2n \frac{dx}{dt} + 2n^2 x = g$$

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A1

4

(b) CF $x = Ae^{pt}$

$$p^2 + 2np + 2n^2 = 0$$

M1

$$p = \frac{-2n \pm \sqrt{4n^2 - 8n^2}}{2}$$

$$= -n \pm ni$$

A1

$$\therefore x = e^{-nt} (A \cos nt + B \sin nt)$$

A1

$$\text{PI } x = \frac{g}{2n^2}$$

B1

$$x = e^{-nt} (A \cos nt + B \sin nt) + \frac{g}{2n^2}$$

M1

When $t = 0$, $x = 0$,

$$A = -\frac{g}{2n^2}$$

B1

$$\frac{dx}{dt} = -ne^{-nt} (A \cos nt + B \sin nt) +$$

M1

$$e^{-nt} (-nA \sin nt + nB \cos nt)$$

When $t = 0$, $\frac{dx}{dt} = 0$

$$\Rightarrow -nA + nB = 0$$

$$B = -\frac{g}{2n^2}$$

A1

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$$x = \frac{g}{2n^2} \{1 - e^{-nt} (\cos nt + \sin nt)\}$$

A1

9

[13]

Q28.

(a) CF $\ddot{x} + \dot{x} = 0$
 $x = Ae^{nt} \quad n^2 + n = 0$

M1

$$n = 0, -1$$

$$x = A + Be^{-t}$$

A1

PI $x = C \cos 2t + D \sin 2t$

Need both terms

M1

$$-4C \cos 2t - 4D \sin 2t - 2C \sin 2t + 2D \cos 2t = k \sin 2t$$

$$-4C + 2D = 0 \text{ and } -4D - 2C = k$$

$$C = -\frac{k}{10}; D = -\frac{k}{5}$$

A1 if at least one correct

A1

$$x = A + Be^{-t} - \frac{k}{10} \cos 2t - \frac{k}{5} \sin 2t$$

ft dep on M2 above

A1ft

$$t = 0, x = a \quad a = A + B - \frac{k}{10}$$

ft dep on M2 above

A1ft

$$\dot{x} = -Be^{-t} + \frac{k}{5} \sin 2t - \frac{2k}{5} \cos 2t$$

M1ft

$$t = 0, \dot{x} = 0; \quad 0 = -B - \frac{2k}{5} \quad B = -\frac{2k}{5}$$

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A1

$$A = a + \frac{k}{10} - B$$

$$A = a + \frac{k}{2}$$

A1

$$x = a + \frac{k}{2} - \frac{2k}{5} e^{-t} - \frac{k}{10} \cos 2t - \frac{k}{5} \sin 2t$$

A1

11

- (b) If e^{-t} term may be ignored, range of $\frac{k}{10} \cos 2t + \frac{2k}{10} \sin 2t$ using

$$a \cos \theta + b \sin \theta = R \cos(\theta - \alpha)$$

M1

$$\text{is } \pm \frac{k\sqrt{5}}{10}$$

Max / min

M1 A1

$\therefore$ values are

ft dep on all M gained in (a)

A1ft

4

$$a + \frac{k}{2} - \frac{k\sqrt{5}}{10} \text{ and } a + \frac{k}{2} - \frac{k\sqrt{5}}{10}$$

if differentiation used, **sc 2** for either

These are $a + 0.276k$ and $a + 0.7236k$

[15]

Q29.

(a)



Using $F = ma$,

$$m\ddot{x} = mg - T - 2mnv$$

M1

$$= mg - 6mn^2 a, \frac{x}{a} - 2mnv$$

A1

$$m\ddot{x} + 6mn^2 x + 2mnv = mg$$

$$\text{i.e. } \frac{d^2x}{dt^2} + 2n\frac{dx}{dt} + 6n^2x = g$$

A1

3

(b) Substituting $x = Ae^{pt}$

M1

$$p^2 + 2np + 6n^2 = 0$$

$$p = \frac{(-2n \pm \sqrt{4n^2 - 24n^2})}{2}$$

$$= -n \pm \sqrt{5} i n$$

A1

$$\therefore \text{CF is } x = e^{-nt} (A \cos \sqrt{5} nt + B \sin \sqrt{5} nt)$$

A1

$$\text{PI is } x = \frac{g}{6n^2}$$

B1

$$\therefore \text{GS is } x = e^{-nt} (A \cos \sqrt{5} nt + B \sin \sqrt{5} nt)$$

$$= \frac{g}{6n^2}$$

A1

$$\text{When } t = 0, x = 0 \Rightarrow 0 = A + \frac{g}{6n^2}$$

M1

$$\therefore A = -\frac{g}{6n^2}$$

A1

$$\text{When } t = 0, \frac{dx}{dt} = 0;$$

$$\frac{dx}{dt} = -ne^{-nt} (A \cos \sqrt{5} nt + B \sin \sqrt{5} nt)$$

$$\left| + e^{-nt} (-\sqrt{5} n A \sin \sqrt{5} nt + \sqrt{5} n B \cos \sqrt{5} nt) \right|$$

$$+ e^{-nt} (-\sqrt{5} n A \sin \sqrt{5} nt + \sqrt{5} n B \cos \sqrt{5} nt)$$

M1

$$\Rightarrow 0 = -n A \times \sqrt{5} n B)$$

$$\therefore B = \frac{1}{\sqrt{5}} A = -\frac{g}{6\sqrt{5}n^2}$$

A1

$$\therefore x = \frac{g}{6n^2} -$$

$$\left(\frac{g}{6n^2} \cos \sqrt{5}nt + \frac{g}{6\sqrt{5}n^2} \sin \sqrt{5}nt \right) e^{-nt}$$

A1

10

(c) $\frac{dx}{dt} =$

$$ne^{-nt} \left(\frac{g}{6n^2} \cos \sqrt{5}nt + \frac{g}{6\sqrt{5}n^2} \sin \sqrt{5}nt \right) -$$

$$e^{-nt} \left(-\frac{\sqrt{5}g}{6n} \sin \sqrt{5}nt + \frac{g}{6n} \cos \sqrt{5}nt \right)$$

M1

$$\frac{dx}{dt} = 0 \text{ when } \cos \sqrt{5}nt + \frac{1}{\sqrt{5}} \sin \sqrt{5}nt +$$

$$\sqrt{5} \sin \sqrt{5}nt \cos \sqrt{5}nt = 0$$

$$\sin \sqrt{5}nt = 0$$

M1A1

First time at rest is when $t = \frac{\pi}{\sqrt{5}n}$

$$\text{When } t = \frac{\pi}{\sqrt{5}n}, \quad \frac{dx}{dt} = 0 \quad \text{SC2}$$

A1

4

[17]

Examiner reports

Q1.

In this less familiar question, more students failed to gain any marks than in any other question on the paper. In general, only the more able students started to look for a particular integral of the form ax^2e^x , having realised that the more 'obvious' forms were included in the given complementary function. Some other students persevered after trying the forms ae^x and then axe^x , showing that each failed before then succeeding with the form ax^2e^x but a significant minority of students instead abandoned their solutions after trying one or both of these incorrect forms.

Q2.

The method to find the general solution of this second order differential equation was well understood by almost all students. The most common error was to incorrectly solve the auxiliary equation to obtain roots $3 \pm 2i$ although many then presented correct follow through work to score 5 of the 6 marks.

Q3.

This question, which tested the solution of a second order differential equation with the form of the particular integral given, was a good source of marks for many students; especially before the boundary conditions in part (c) became part of the test. The majority of students found the correct values for the constants a , b and c in part (a) but sign errors were sometimes seen; the most common being to equate $-4c$ to 8 instead of -8 from consideration of the coefficients of e^{-3x} . In part (b), most students showed that they knew how to find the complementary function and then the general solution. Part (c) caused students more problems than the previous two parts. Although most students scored the mark for correctly dealing with the given boundary condition at $x = 0$, a significant minority did not realise that the coefficient of e^x , in the expression for $\frac{dy}{dx}$, had to be 0 for the boundary condition $\frac{dy}{dx} \rightarrow -1$ as $x \rightarrow \infty$ to be satisfied. As mentioned in previous reports, the examiners expect students to give special attention to certain limits that are within the specification. In this question, the limit of xe^{-3x} as $x \rightarrow \infty$ needed to be treated in isolation, and this was not considered by a significant number of students.

Q4.

The majority of candidates scored full marks for solving this second order differential equation, subject to the given boundary conditions, correctly. Other than arithmetic and algebraic slips, the main source of misunderstanding was to give the complementary function as $Ae^{-x}(\cos 3x + \sin 3x)$, which lead to a general solution of a second order differential equation with only one arbitrary constant.

Q5.

In part (a), those candidates who started by writing $\frac{du}{dx} = y + x \frac{dy}{dx}$ and then moved on to find $\frac{d^2u}{dx^2}$ generally presented a convincing solution to show that the given substitution transformed one differential equation into the other.

The above average candidates generally showed that they had a thorough understanding of the method required to find the general solution of the differential equation although a few marks were lost through careless slips, most notably, due to ' $9(px + q) = 9px + q$ '.

Three more serious errors / lack of understanding seen, which led to a greater loss of marks, are illustrated by (I) $y = e^{-3x}(Ax + B)$ or (II) $y = e^{-3u}(Au + B)$ or (III) $u = Ae^{-3x}$ as

the CF for $\frac{d^2u}{dx^2} + 6\frac{du}{dx} + 9u = 18x$.

Q6.

Most candidates were able to find the correct values of the constants p and q so that the given expression was a particular integral of the differential equation. An even higher proportion of candidates scored full marks for their general solution in part (b) as examiners allowed follow through on incorrect values for p and q . However, less than 20% of the candidates scored full marks for their answers to part (c). This was almost entirely due to no explicit consideration being given to the value of xe^{-2x} as $x \rightarrow \infty$. More surprisingly, a significant minority of candidates could not correctly deal with the boundary

condition $\frac{dy}{dx} \rightarrow 0$ as $x \rightarrow \infty$ as they had an e^x term in their final answer.

Q7.

In part (a), it was pleasing to see a higher proportion of candidates than in previous papers using the given form of the particular integral rather than introducing the extra term qe^{-2x} . The majority of candidates used the product rule correctly to find the value of the constant p . Most candidates showed that they knew the methods required to solve the second order differential equation, although a minority of candidates made the error of applying the boundary conditions to the complementary function before adding on the particular integral.

Q8.

This unstructured question was answered well by many of the candidates. However, a significant minority of candidates did not even write down the correct form of the auxiliary equation, or they solved it incorrectly to give real rather than imaginary values. When finding the particular integral, a large number of candidates considered more terms than they needed to, and they did not always go on to show that the relevant coefficients of the extra terms were zero. The majority of candidates knew that they had to find a complementary function and a particular integral and almost all scored the final mark for combining the two to give the general solution.

Q9.

This was another very good source of marks for candidates, with many fully correct solutions presented. In part (a) some candidates decided to ignore the given form of the particular integral and worked with $p\cos 2x + q\sin 2x$. Such an approach was not penalised

by examiners provided the candidate showed that both $p = 0$ and $q = -\frac{1}{3}$. The vast majority of candidates showed that they knew the methods to solve the second order differential equation but errors in forming and solving the auxiliary equation were sources of loss of marks. Real solutions from the auxiliary equation were more heavily penalised than other errors.

Q10.

In part (a), many candidates decided to ignore the given form of the particular integral and worked with $a \cos x + b \sin x$. Such an approach was not penalised by examiners provided the candidate showed that both $a = 0$ and $b = 2$. The vast majority of candidates showed that they knew the methods required to solve the second order differential equation, but arithmetical errors in solving the auxiliary equation $m^2 + 2m + 5 = 0$ or in

applying the boundary condition $\frac{dy}{dx} = 4$ when $x = 0$ or the wrong differentiation of $\cos 2x$ were sources of loss of marks as well as the more serious error of applying the boundary conditions to the complementary function before adding on the particular integral.

Q11.

This question, which required candidates to solve a second order differential equation, was generally a good source of marks. It was disappointing to see some candidates trying to solve the auxiliary equation, $m^2 + 4m + 5 = 0$, by factorisation. They obtained real solutions and this error was penalised heavily. Better candidates were able to write down the correct complementary function and find the particular integral but some wasted valuable time by starting with $y_p = ax^2 + bx + c$ and showing that both a and b were zero. Candidates who were able to find the correct general solution in part (a) usually went on to apply the given boundary conditions correctly in their answers to part (b).

Q12.

In part (a), many candidates scored high marks for finding the general solution of the second order differential equation, with the most common slip being a wrong expansion of brackets, which led to -9 instead of -3 for the constant part of the particular integral. A more serious but less common error was to look for a particular integral of the form $axe^{-2x} + b$. Part (b) proved, as expected, to be more of a challenge to candidates, with many unable to deal with the boundary condition expressed as a limit. This was the first time that this had appeared on an examination paper for this unit and, in general, only the better candidates could handle it.

Q13.

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This question, on finding the general solution of a second order differential equation, was a good source of marks for many candidates. Some candidates failed to write their final answer in the form $y = f(x)$, but a more common error was not finding/stating “2” as part of the particular integral.

Q14.

Many candidates gained very high marks for this opening question which tested the solution of a second order differential equation. In part (a), although the form of the particular integral was given in the question, a significant minority of candidates preferred to discover it themselves and tried particular integrals $y = ae^{5x}$ and $y = axe^{5x}$, showing that each failed before trying $y = ax^2e^{5x}$ and finding that $a = 3$. These candidates were not penalised for this approach but clearly they used up valuable time in doing more than what was required. Some candidates also found the complementary function in part (a) and full relevant credit was given even if this work was not repeated in part (b). Most candidates indicated that they knew how to find the complementary function and how to use it to find the general solution but a significant number failed to score full marks because they did not give their final answer in the form ‘ $y = f(x)$ ’ where $f(x)$ contains two

arbitrary constants.

Q15.

The majority of candidates gained at least ten of the twelve available marks for this opening question which tested the solution of a second order differential equation. Part (a) caused few problems although there were a few who had difficulty simplifying

$$\frac{\sqrt{4-8}}{2}$$

correctly. The most common method error in the question occurred in part(b)(i). These candidates considered a particular integral of the form ' ax ' instead of the correct form ' $ax + b$ '. In general, candidates were able to apply the given boundary conditions in a correct manner.

Q16.

Most candidates gained very high marks for this opening question which tested the solution of a second order differential equation. In part (a) it was acceptable to differentiate the given result and substitute into the given differential equation. Although sign errors were seen, the vast majority who used this approach scored all three marks. Those who took a general form for the particular integral sometimes failed to find all the unknown constants.

Part (b) was generally answered correctly although some gave the final answer as $y = Ae^{4x} + Be^{2x} + 8x - 10 - 10\cos 2x$. Except for some careless slips in using $y'(0) = 0$ the final part of the question was answered well.

Q17.

Part (a) was done well by many students, but some struggled to provide a convincing justification for the differential equation. A clear statement of the resultant force and then a clear application of Newton's Second law would have helped some students.

The majority of students did very well with the solution of the differential equation and seemed to be comfortable with this part of the question. The most common error seen was probably to use $\dot{x} = 0$ rather than $\dot{x} = 12.5$. There were however a whole variety of minor arithmetical and algebraic errors. Part (c) was done well but obviously some students were using incorrect expressions for the speed.

Q18.

This was another popular question for many, although a few failed to appreciate the vertical nature of the motion, falling down in part (b)(i). Parts (a) and (b)(ii) and (iii) were mostly successful but a few could not provide a correct formula for the maximum speed for this motion.

Q19.

This question proved challenging for all candidates. The main difficulty was in appreciating the focus of the question being on the motion of the particle P , despite the hint in the early parts of the question. Part (a) was poorly answered, showing scant knowledge of the implications of forced motion. Despite this, part (b) was answered quite well, with some candidates realising that they needed to differentiate the answer from part (a). Part (c) was rarely answered well, due to reasons already mentioned, although many faked this solution. Most candidates were able to attempt the solution to the differential equation in part (d), but there were a variety of errors; some giving non-real solutions to the auxiliary

equation, others ignoring the particular integral, or evaluating constants before including this.

Q20.

Part (a) was completed successfully by most candidates. Part (b) was often successfully answered, with a common error being the omission of the weight and an incorrect expression for the tension in the general position; often these two errors cancelled each other out in subsequent working, leading to an apparently correct solution. The main error in part (c) was to stop work having found the full general solution and not to evaluate the remaining constants. Part (d) was not answered well.

Q21.

Part (a) was answered well, although again some candidates were clearly working back from the printed result to obtain the initial equation of motion. Part (b) was attempted well while the common error was in using $x = a$ when $t = 0$, not noticing that x was the distance below A , and initially the particle was released at A ; thus $\alpha = 0$ when $t = 0$. Relatively few candidates were totally accurate in finding the velocity of the particle in part (c) and only these successful candidates could notice that the terms in $\cos \sqrt{3\pi t}$ cancelled out, leaving only the terms containing $\sin \sqrt{3\pi t}$.

Q22.

The great majority of candidates were able to use Hooke's law and Newton's second law to answer parts (a) and (b) of this question. Some candidates were unable to explain why $\dot{x} = 1 - v$ for part (c)(i) but were still able to gain full marks for part (c)(ii).

Some candidates were able to answer part (d) of the question without answering part (c). Even though only the general solution for the differential equation was requested, some candidates proceeded to find the particular solution also. Many candidates were able to state the limiting value of x requested in part (e).

Q23.

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A relatively large number of candidates had difficulty drawing the force/displacement diagram requested for part (a) of this question. A great majority of the candidates were able to answer part (b) correctly even when some of these candidates had not drawn a correct diagram.

The responses to part (c) were generally good, indicating a high level of knowledge and skills on the part of candidates in solving second order differential equations where the roots of the auxiliary equation are complex.

Part (d) of this question proved too difficult for some candidates.

Q25.

It was pleasing to see good responses from the great majority of candidates to this question. Candidates showed a high degree of competence and confidence in solving the second order differential equation and in interpreting their solutions to part (c).

Q26.

This question was answered well. Apart from a few arithmetical errors most candidates

obtained correctly $x = (18\cos t + 51\sin t)e^{-3t} + \sin 3t - 18\cos 3t$. Many candidates appreciated that, when t becomes large, the e^{-3t} term became negligible but only a few described fully, as required, the resulting expression for x , $\sin 3t - 18\cos 3t$.

Q27.

In part (a), a significant proportion of candidates worked back from the given result. Others used x as the position beyond the equilibrium position rather than as stated, and then had to 'lose' terms. In part (b), candidates made very good progress with many obtaining the correct result. The common error was forgetting an "n" in the

differentiation of one of the terms when finding $\frac{dx}{dt}$ to insert a boundary condition.

Other candidates could not simplify correctly the term $\frac{-2n \pm \sqrt{-4n^2}}{2}$.

Q28.

The complementary function $x = A + Be^{-t}$ caused difficulty to some candidates after they had obtained $n = 0$ and $n = -1$ from their initial substitution of $x = Ae^{nt}$. A number then wrote $x = (A + Bt)e^{-t}$.

The particular integral $x = C\cos 2t + D\sin 2t$ was also often simplified to $x = C\sin 2t$, with candidates ignoring the $\cos 2t$ term that they had obtained in their differentiation. Those who used the correct form of the complementary function and the correct form of the particular integral usually obtained the correct general solution.

After making the e^{-t} term go to zero in part (b), some used differentiation to find the maximum and minimum values but this was very time consuming and offered many opportunities for making errors.

Better candidates appreciated that the range in values of

$\frac{k}{10}\cos 2t + \frac{k}{5}\sin 2t$ was $\pm \sqrt{\left(\frac{k}{10}\right)^2 + \left(\frac{k}{5}\right)^2}$ from the use of $a\cos\theta + b\sin\theta = R\cos(\theta - \alpha)$.

Q29.

Many candidates attempted to find the extension of the string in the equilibrium position, although in this question it was not relevant. These candidates had no mg force present when they found their differential equation. As this equation was printed on the question paper, candidates who then 'created' the mg force rarely scored marks in part (a). In part (b), candidates usually obtained the general solution $x = e^{-nt}(A\cos\sqrt{5}nt + B\sin\sqrt{5}nt)$, and most attempted to find the particular solution. They then showed the correct method for obtaining the actual solution. Those who correctly found the answer to part (b) often

substituted $t = \frac{\pi}{\sqrt{5}n}$ to show that the particle was at rest.

Only the better candidates differentiated their answer to show that the particle was at rest.